

Pattern Formation in Neural Fields: From Bumps to Collective Decisions

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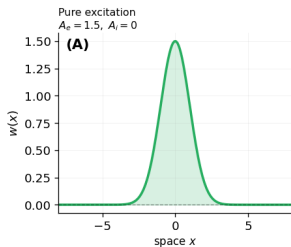
ISRC-CN3 Summer School · 19 August 2026

Amari neural field equation

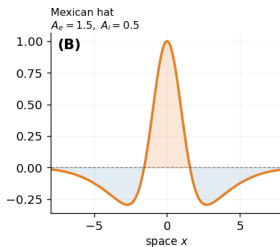
$$\tau \partial_t u(x, t) + u(x, t) = \int_{\mathbb{R}} w(x-y) f(u(y, t)) dy, \quad f(u) = \frac{1}{2} (1 + \tanh \beta(u - \eta))$$

Difference-of-Gaussians (Mexican-hat) kernel

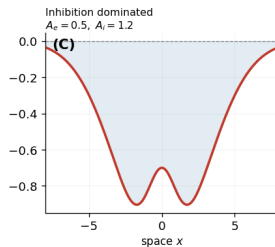
$$w(x) = A_e e^{-x^2/2\sigma_e^2} - A_i e^{-x^2/2\sigma_i^2} \quad (\text{local excitation, lateral inhibition})$$



Net drive: $\dot{w}(0) = +3.76 > 0$
Only excitatory coupling.
→ Homogeneous high-activity state.



Net drive: $\dot{w}(0) = 0$ (E/I balance)
Local excit., lateral inhib.
→ Turing instability / bump pattern.



Net drive: $\dot{w}(0) = -7.77 < 0$
Inhibition suppresses activity.
→ Stable low-activity resting state.

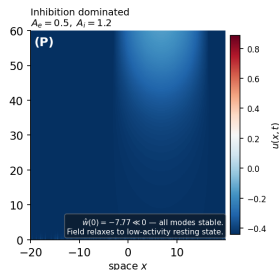
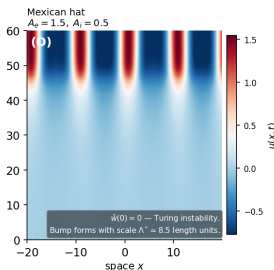
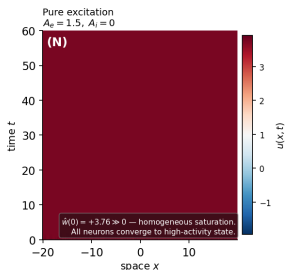
Stability in one line: which spatial scales grow?

Perturb the homogeneous state u^* mode by mode, $u = u^* + \epsilon e^{\lambda t} e^{ikx}$:

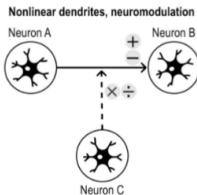
$$\lambda(k) = \frac{f'(u^*) \hat{w}(k) - 1}{\tau}$$

$\hat{w}$ real, even $\Rightarrow \lambda(k) \in \mathbb{R}$: *stationary*, no travelling

Turing instability: $\lambda(0) < 0$ but $\lambda(k^*) > 0$ for some $k^* \neq 0$ — the connectivity, not the input, selects the scale $\Lambda^* = 2\pi/k^* \approx 8.5$.



Nonlinear opinion dynamics



Attention u_0 plays the role of *gain*;
modulation m_{ijk} is a *triadic* interaction —
as when a third neuron (or glia) regulates
a synapse.

NOD model with modulation
term (Wilson–Cowan/Hopfield
structure) (Moreno-Bizyaeva et al
2024)

$$\tau \dot{x}_i = -x_i + b_i + S\left(\sum_j a_{ij}(u_0 + \sum_k m_{ijk} x_k^n) x_j\right)$$

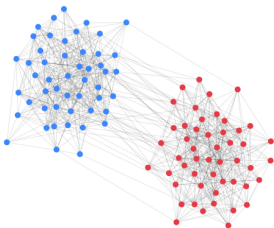
Today: $b = 0$, $M = 0$,
degree-normalised

$$\tau \dot{\mathbf{x}} = -\mathbf{x} + \tanh(u_0 W \mathbf{x}), \quad W = D^{-1} A$$

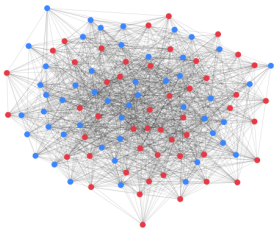
Modulation is invisible at the bifurcation ($J(0)$
independent of M) — it shapes what happens *after*.

100 agents. Consensus and polarization

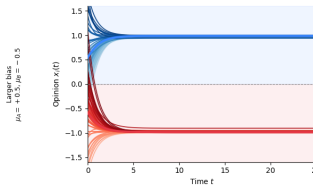
Panel A graph (sparse, $p_{out} = 0.02$)



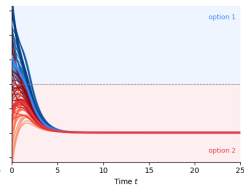
Panel B/C graph (well-mixed, $p_{out} = 0.26$)



Polarization graph ($p_{out} = 0.02$)



Consensus graph ($p_{out} = 0.26$)



- Normalisation pins $\lambda_{\max}(W) = 1 \Rightarrow$ universal threshold $u_0^* = 1$.
- Sparse inter-community coupling: **polarisation**; dense: **consensus** — controlled only by $\rho = p_{out}/p_{in}$.

Main contribution: the mean-field limit of NOD

Opinion distribution $f(x, t)$: nonlinear nonlocal continuity equation

$$\begin{aligned}\partial_t f + \nabla_x (V[f] f) &= 0, & V[f](x) &= \frac{1}{\tau} \left(-x + b + S(a(u_0 + m \mu_n[f]) \mu_1[f]) \right) \\ \mu_1[f] &= \int y f(y, t) dy \quad (\text{mean opinion}), & \mu_n[f] &= \int y^n f(y, t) dy \quad (n\text{-th moment})\end{aligned}$$

Biological reading: f is a *population density over activation states*. Each unit feels the rest only through two summary statistics: the mean drive μ_1 , and μ_n multiplying the attention — *global, state-dependent gain modulation*: the population regulating its own gain, as broadcast neuromodulation does.

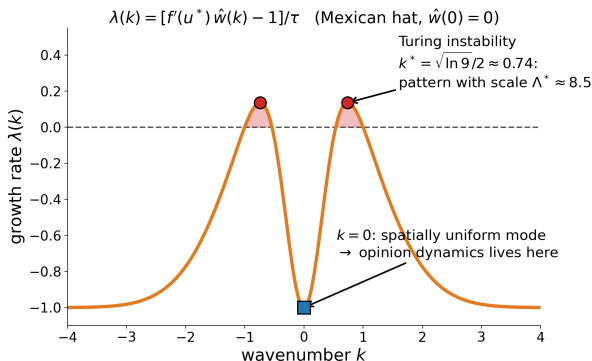
Same bifurcation, no space

One formula: growth rate = (gain $\times$ coupling eigenvalue $- 1$)/ τ

$$\text{NOD: } \frac{u_0 \lambda_k - 1}{\tau}$$

$$\text{NFE: } \frac{f'(u^*) \hat{w}(k) - 1}{\tau}$$

eigenvectors of $W \longleftrightarrow$ Fourier modes of w *



Open question: what is the discrete analogue of a Mexican hat — which network structure would let agents form genuinely *spatial* opinion patterns?

Backup: derivation via the empirical measure

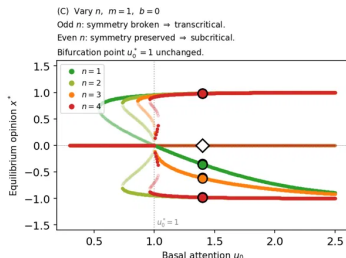
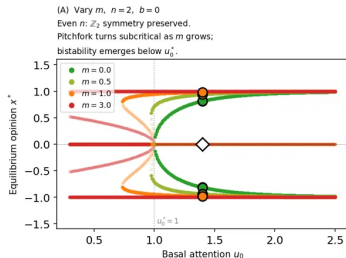
Empirical measure $\mu_t^N = \frac{1}{N} \sum_{i=1}^N \delta_{x_i(t)}$; testing against $\varphi \in C_c^\infty$ and passing $N \rightarrow \infty$ yields the weak formulation

$$\frac{d}{dt} \int \varphi d\mu_t = \int V[\mu_t] \varphi' d\mu_t.$$

- The velocity is *affine* in x ; all nonlinearity is concentrated in the two scalar functionals μ_1, μ_n .
- All-to-all (exchangeable) coupling: the graph structure is averaged out — consistent with the $k = 0$ picture.

Note: this is the continuum limit of the **opinion** model — it is *not* the Amari equation; the two frameworks are connected through the shared bifurcation structure.

Backup: modulation shapes the bifurcation, not the threshold



- Threshold $u_0^* = 1$ **unchanged** by m, n (Jacobian at 0 independent of M).
- **Parity of n decides the type:**
even n preserves $\mathbb{Z}_2$ symmetry $\Rightarrow$ pitchfork (subcritical as m grows: bistability below threshold); odd n breaks it $\Rightarrow$ transcritical.
- Lyapunov–Schmidt reduction on the consensus mode.